7 Electrodynamics

7.1 Electromotive Force

7.1.1 Ohm's Law

To make a current flow, you have to *push* on the charges. How *fast* they move, in response to a given push, depends on the nature of the material. For most substances, the current density **J** is proportional to the *force per unit charge*, **f**:

$$\mathbf{J} = \sigma \mathbf{f}. \tag{7.1}$$

The proportionality factor σ (not to be confused with surface charge) is an empirical constant that varies from one material to another; it's called the **conductivity** of the medium. Actually, the handbooks usually list the *reciprocal* of σ, called the **resistivity**: $\rho = 1/\sigma$ (not to be confused with charge density – I'm sorry, but we're running out of Greek letters, and this is the standard notation). Some typical values are listed in Table 7.1. Notice that even *insulators* conduct slightly, though the conductivity of a metal is astronomically greater; in fact, for most purposes metals can be regarded as **perfect conductors**, with $\sigma = \infty$, while for insulators we can pretend $\sigma = 0$.

In principle, the force that drives the charges to produce the current could be anything – chemical, gravitational, or trained ants with tiny harnesses. For *our* purposes, though, it's usually an electromagnetic force that does the job. In this case Eq. 7.1 becomes

$$\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \tag{7.2}$$

Ordinarily, the velocity of the charges is sufficiently small that the second term can be ignored:

$$\boxed{\mathbf{J} = \sigma \mathbf{E}.} \tag{7.3}$$

(However, in plasmas, for instance, the magnetic contribution to **f** can be significant.) Equation 7.3 is called **Ohm's law**, though the physics behind it is really contained in Eq. 7.1, of which 7.3 is just a special case.

I know: you're confused because I said $\mathbf{E} = \mathbf{0}$ inside a conductor (Section 2.5.1). But that's for *stationary* charges ($\mathbf{J} = \mathbf{0}$). Moreover, for *perfect* conductors $\mathbf{E} = \mathbf{J}/\sigma = \mathbf{0}$ even if current *is* flowing. In practice, metals are such good conductors that the electric field required to drive current in

Table 7.1. Resistivities, in ohm meters (all values are for 1 atm, 20°C). Data from *Handbook of Chemistry and Physics*, 91st ed., CRC Press, Boca Raton (2010) and other references.

Material	Resistivity	Material	Resistivity
Conductors:		*Semiconductors:*	
Silver	1.59×10^{-8}	Sea water	0.2
Copper	1.68×10^{-8}	Germanium	0.46
Gold	2.21×10^{-8}	Diamond	2.7
Aluminum	2.65×10^{-8}	Silicon	2500
Iron	9.61×10^{-8}	*Insulators:*	
Mercury	9.61×10^{-7}	Water (pure)	8.3×10^{3}
Nichrome	1.08×10^{-6}	Glass	10^{9}–10^{14}
Manganese	1.44×10^{-6}	Rubber	10^{13}–10^{15}
Graphite	1.6×10^{-5}	Teflon	10^{22}–10^{24}

them is negligible. Thus we routinely treat the connecting wires in electric circuits (for example) as equipotentials. **Resistors**, by contrast, are made from *poorly* conducting materials.

Example 7.1. A cylindrical resistor of cross-sectional area A and length L is made from material with conductivity σ. (See Fig. 7.1; as indicated, the cross section need not be circular, but I *do* assume it is the same all the way down.) If we stipulate that the potential is constant over each end, and the potential difference between the ends is V, what current flows?

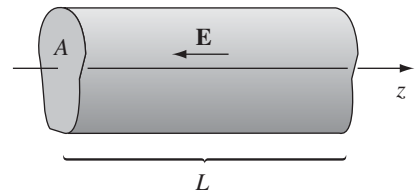


Fig. 7.1

Solution. As it turns out, the electric field is *uniform* within the wire (I'll prove this in a moment). It follows from Eq. 7.3 that the current density is also uniform, so

$$I = JA = \sigma EA = \frac{\sigma A}{L} V.$$

Example 7.2. Two long coaxial metal cylinders (radii a and b) are separated by material of conductivity σ (Fig. 7.2). If they are maintained at a potential difference V, what current flows from one to the other, in a length L?

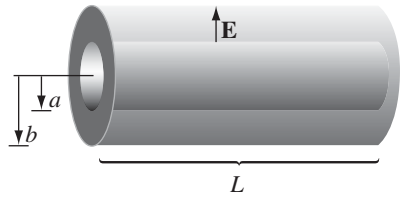


Fig. 7.2

Solution. The field between the cylinders is

$$\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 s}\hat{\mathbf{s}},$$

where λ is the charge per unit length on the inner cylinder. The current is therefore

$$I = \int \mathbf{J} \cdot d\mathbf{a} = \sigma \int \mathbf{E} \cdot d\mathbf{a} = \frac{\sigma}{\epsilon_0}\lambda L.$$

(The integral is over any surface enclosing the inner cylinder.) Meanwhile, the potential difference between the cylinders is

$$V = -\int_b^a \mathbf{E} \cdot d\mathbf{l} = \frac{\lambda}{2\pi\epsilon_0}\ln\left(\frac{b}{a}\right),$$

so

$$I = \frac{2\pi\sigma L}{\ln(b/a)}V.$$

As these examples illustrate, the total current flowing from one **electrode** to the other is proportional to the potential difference between them:

$$\boxed{V = IR.} \tag{7.4}$$

This, of course, is the more familiar version of Ohm's law. The constant of proportionality R is called the **resistance**; it's a function of the geometry of the arrangement and the conductivity of the medium between the electrodes. (In Ex. 7.1, $R = (L/\sigma A)$; in Ex. 7.2, $R = \ln(b/a)/2\pi\sigma L$.) Resistance is measured in **ohms** (Ω): an ohm is a volt per ampere. Notice that the proportionality between V and I is a direct consequence of Eq. 7.3: if you want to double V, you simply double the charge on the electrodes – that doubles $\mathbf{E}$, which (for an ohmic material) doubles $\mathbf{J}$, which doubles I.

For *steady* currents and *uniform* conductivity,

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \frac{1}{\sigma}\boldsymbol{\nabla} \cdot \mathbf{J} = 0 \tag{7.5}$$

(Eq. 5.33), and therefore the charge density is zero; any unbalanced charge resides on the surface. (We proved this long ago, for the case of *stationary* charges, using the fact that $\mathbf{E} = \mathbf{0}$; turns out, it is still true when steady currents flow, for media satisfying Eq. 7.3.) It follows, in particular, that Laplace's equation holds within a homogeneous ohmic material carrying

a steady current, so all the tools and tricks of Chapter 3 are available for calculating the potential.

Example 7.3. I asserted that the field in Ex. 7.1 is *uniform*. Let's prove it.

Solution. Within the cylinder V obeys Laplace's equation. What are the boundary conditions? At the left-hand end the potential is constant – we may as well set it equal to zero. At the right-hand end the potential is likewise constant – call it V_0. On the cylindrical surface, $\mathbf{J} \cdot \hat{\mathbf{n}} = 0$, or else charge would be leaking out into the surrounding space (which we take to be nonconducting). Therefore $\mathbf{E} \cdot \hat{\mathbf{n}} = 0$, and hence $\partial V/\partial n = 0$. With V or its normal derivative specified on all surfaces, the field is uniquely determined (Prob. 3.5). But it's easy to guess *one* potential that obeys Laplace's equation and fits these boundary conditions:

$$V(z) = \frac{V_0 z}{L},$$

where z is measured along the axis. The uniqueness theorem guarantees that this is *the* solution. The corresponding field is

$$\mathbf{E} = -\boldsymbol{\nabla} V = -\frac{V_0}{L}\hat{\mathbf{z}},$$

which is indeed uniform.

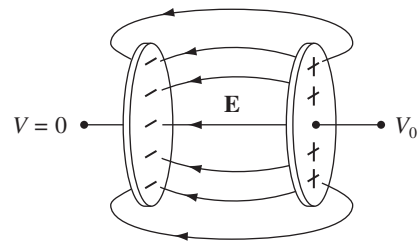


Fig. 7.3

Contrast the enormously more difficult problem that arises if the conducting material is removed, leaving only a metal plate at either end (Fig. 7.3). Evidently, in the present case charge arranges itself over the surface of the wire in just such a way as to produce a nice uniform field within.[1]

I don't suppose there is any formula in physics more familiar than Ohm's law, and yet it's not really a true law, in the sense of Coulomb's or Ampère's; rather, it is a "rule of thumb" that applies pretty well to many substances. You're not going to win a Nobel prize for finding an exception. In fact, when you stop to think about it, it's a little surprising that Ohm's law *ever* holds.

[1] *Calculating* this surface charge is not easy. See, for example, J. D. Jackson, *Am. J. Phys.* **64**, 855 (1996). Nor is it a simple matter to determine the field *outside* the wire – see Prob. 7.45.

After all, a given field **E** produces a force $q\mathbf{E}$ (on a charge q), and according to Newton's second law, the charge will accelerate. But if the charges are *accelerating*, why doesn't the current *increase* with time, growing larger and larger the longer you leave the field on? Ohm's law implies, on the contrary, that a constant field produces a *constant* current, which suggests a constant *velocity*. Doesn't that contradict Newton's law?

No, for we are forgetting the frequent collisions electrons make as they pass down the wire. It's a little like this: Suppose you're driving down a street with a stop sign at every intersection, so that, although you accelerate constantly in between, you are obliged to start all over again with each new block. Your *average* speed is then a constant, in spite of the fact that (save for the periodic abrupt stops) you are always accelerating. If the length of a block is λ and your acceleration is a, the time it takes to go a block is

$$t = \sqrt{\frac{2\lambda}{a}},$$

and hence your average velocity is

$$v_{\text{ave}} = \frac{1}{2}at = \sqrt{\frac{\lambda a}{2}}.$$

But wait! That's no good *either*! It says that the velocity is proportional to the *square root* of the acceleration, and therefore that the current should be proportional to the square root of the field! There's another twist to the story: In practice, the charges are already moving very fast because of their thermal energy. But the thermal velocities have random directions, and average to zero. The **drift velocity** we are concerned with is a tiny extra bit (Prob. 5.21). So the time between collisions is actually much shorter than we supposed; if we assume for the sake of argument that all charges travel the same distance λ between collisions, then

$$t = \frac{\lambda}{v_{\text{thermal}}},$$

and therefore

$$v_{\text{ave}} = \frac{1}{2}at = \frac{a\lambda}{2v_{\text{thermal}}}.$$

If there are n molecules per unit volume, and f free electrons per molecule, each with charge q and mass m, the current density is

$$\mathbf{J} = nfq\mathbf{v}_{\text{ave}} = \frac{nfq\lambda}{2v_{\text{thermal}}}\frac{\mathbf{F}}{m} = \Big(\frac{nf\lambda q^2}{2mv_{\text{thermal}}}\Big)\mathbf{E}. \tag{7.6}$$

I don't claim that the term in parentheses is an accurate formula for the conductivity,[2] but it does indicate the basic ingredients, and it correctly predicts that conductivity is proportional to the density of the moving charges and (ordinarily) decreases with increasing temperature.

[2] This classical model (due to Drude) bears little resemblance to the modern quantum theory of conductivity. See, for instance, D. Park's *Introduction to the Quantum Theory*, 3rd ed., McGraw-Hill, New York (1992), Chap. 15.

As a result of all the collisions, the work done by the electrical force is converted into heat in the resistor. Since the work done per unit charge is V and the charge flowing per unit time is I, the power delivered is

$$\boxed{P = VI = I^2 R.} \tag{7.7}$$

This is the **Joule heating law**. With I in amperes and R in ohms, P comes out in watts (joules per second).

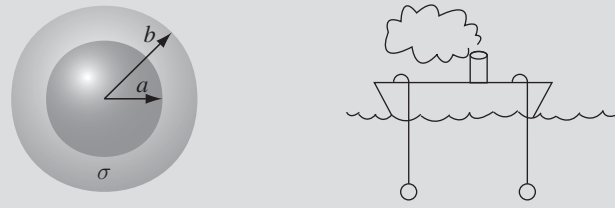


Fig. 7.4

Problem 7.1. Two concentric metal spherical shells, of radius a and b, respectively, are separated by weakly conducting material of conductivity σ (Fig. 7.4a).

(a) If they are maintained at a potential difference V, what current flows from one to the other?
(b) What is the resistance between the shells?
(c) Notice that if $b \gg a$ the outer radius (b) is irrelevant. How do you account for that? Exploit this observation to determine the current flowing between two metal spheres, each of radius a, immersed deep in the sea and held quite far apart (Fig. 7.4b), if the potential difference between them is V. (This arrangement can be used to measure the conductivity of sea water.)

Problem 7.2. A capacitor C has been charged up to potential V_0; at time $t = 0$, it is connected to a resistor R, and begins to discharge (Fig. 7.5a).

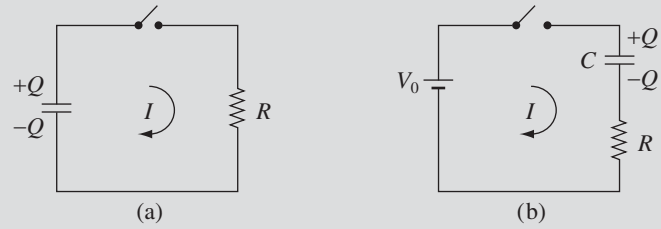


Fig. 7.5

(a) Determine the charge on the capacitor as a function of time, $Q(t)$. What is the current through the resistor, $I(t)$?
(b) What was the original energy stored in the capacitor (Eq. 2.55)? By integrating Eq. 7.7, confirm that the heat delivered to the resistor is equal to the energy lost by the capacitor.

Now imagine *charging up* the capacitor, by connecting it (and the resistor) to a battery of voltage V_0, at time $t = 0$ (Fig. 7.5b).

(c) Again, determine $Q(t)$ and $I(t)$.

(d) Find the total energy output of the battery ($\int V_0 I\, dt$). Determine the heat delivered to the resistor. What is the final energy stored in the capacitor? What fraction of the work done by the battery shows up as energy in the capacitor? [Notice that the answer is independent of R!]

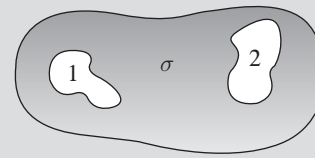


Fig. 7.6

Problem 7.3.

(a) Two metal objects are embedded in weakly conducting (nonpolarizable) material of conductivity σ (Fig. 7.6). Show that the resistance between them is related to the capacitance of the arrangement by

$$R = \frac{\epsilon_0}{\sigma C}.$$

(b) Suppose you connected a battery between 1 and 2, and charged them up to a potential difference V_0. If you then disconnect the battery, the charge will gradually leak off. Show that $V(t) = V_0 e^{-t/\tau}$, and find the **time constant**, τ, in terms of ϵ_0 and σ.

Problem 7.4. Suppose the conductivity of the material separating the cylinders in Ex. 7.2 is not uniform; specifically, $\sigma(s) = k/s$, for some constant k. Find the resistance between the cylinders. [*Hint:* Because σ is a function of position, Eq. 7.5 does not hold, the charge density is not zero in the resistive medium, and $\mathbf{E}$ does not go like $1/s$. But we *do* know that for steady currents I is the same across each cylindrical surface. Take it from there.]

7.1.2 Electromotive Force

If you think about a typical electric circuit – a battery hooked up to a light bulb, say (Fig. 7.7) – a perplexing question arises: In practice, the *current is the same all the way around the loop*; why is this the case, when the only obvious driving force is inside the battery? Off hand, you might expect a large current in the battery and none at all in the lamp. Who's doing the pushing, in the rest of the circuit, and how does it happen that this push is exactly right to produce the same current in each segment? What's more, given that the charges in a typical wire move (literally) at a *snail's* pace (see Prob. 5.21), why doesn't it take half an hour for the current to reach the light bulb? How do all the charges know to start moving at the same instant?

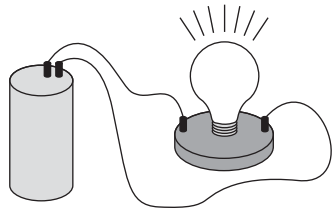


Fig. 7.7

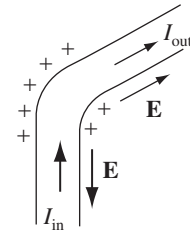


Fig. 7.8

Answer: If the current were *not* the same all the way around (for instance, during the first split second after the switch is closed), then charge would be piling up somewhere, and – here's the crucial point – the electric field of this accumulating charge is in such a direction as to even out the flow. Suppose, for instance, that the current *into* the bend in Fig. 7.8 is greater than the current *out*. Then charge piles up at the "knee," and this produces a field aiming *away* from the kink.[3] This field *opposes* the current flowing in (slowing it down) and *promotes* the current flowing out (speeding it up) until these currents are equal, at which point there is no further accumulation of charge, and equilibrium is established. It's a beautiful system, automatically self-correcting to keep the current uniform, and it does it all so quickly that, in practice, you can safely assume the current is the same all around the circuit, even in systems that oscillate at radio frequencies.

There are really *two* forces involved in driving charge around a circuit: the *source*, $q\mathbf{f}_s$, which is ordinarily confined to one portion of the loop (a battery, say), and an *electrostatic* force, $q\mathbf{E}$, which serves to smooth out the flow and communicate the influence of the source to distant parts of the circuit:

$$\mathbf{f} = \mathbf{f}_s + \mathbf{E}. \tag{7.8}$$

The physical agency responsible for $\mathbf{f}_s$ can be many different things: in a battery it's a chemical force; in a piezoelectric crystal, mechanical pressure is converted into an electrical impulse; in a thermocouple it's a temperature gradient that does the job; in a photoelectric cell it's light; and in a Van de Graaff generator the electrons are literally loaded onto a conveyer belt and swept along. Whatever the *mechanism*, its net effect is determined by the line integral of $\mathbf{f}$ around the circuit:

$$\boxed{\mathcal{E} \equiv \oint \mathbf{f} \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l}.} \tag{7.9}$$

(Because $\oint \mathbf{E} \cdot d\mathbf{l} = 0$ for electrostatic fields, it doesn't matter whether you use $\mathbf{f}$ or $\mathbf{f}_s$.) The quantity $\mathcal{E}$ is called the **electromotive force**, or **emf**, of the circuit. It's a lousy term, since this is not a *force* at all – it's the *integral* of a *force per unit charge*. Some people prefer the word **electromotance**, but emf is so established that I think we'd better stick with it.

[3] The amount of charge involved is surprisingly small – see W. G. V. Rosser, *Am. J. Phys.* **38**, 265 (1970); nevertheless, the resulting field can be detected experimentally – see R. Jacobs, A. de Salazar, and A. Nassar, *Am. J. Phys.* **78**, 1432 (2010).

Within an ideal source of emf (a resistanceless battery,[4] for instance), the *net* force on the charges is *zero* (Eq. 7.1 with $\sigma = \infty$), so $\mathbf{E} = -\mathbf{f}_s$. The potential difference between the terminals (a and b) is therefore

$$V = -\int_a^b \mathbf{E} \cdot d\mathbf{l} = \int_a^b \mathbf{f}_s \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l} = \mathcal{E} \tag{7.10}$$

(we can extend the integral to the entire loop because $\mathbf{f}_s = \mathbf{0}$ outside the source). The function of a battery, then, is to establish and maintain a voltage difference equal to the electromotive force (a 6 V battery, for example, holds the positive terminal 6 V above the negative terminal). The resulting electrostatic field drives current around the rest of the circuit (notice, however, that *inside* the battery $\mathbf{f}_s$ drives current in the direction *opposite* to $\mathbf{E}$).[5]

Because it's the line integral of $\mathbf{f}_s$, $\mathcal{E}$ can be interpreted as the *work done per unit charge*, by the source – indeed, in some books electromotive force is *defined* this way. However, as you'll see in the next section, there is some subtlety involved in this interpretation, so I prefer Eq. 7.9.

Problem 7.5. A battery of emf $\mathcal{E}$ and internal resistance r is hooked up to a variable "load" resistance, R. If you want to deliver the maximum possible *power* to the load, what resistance R should you choose? (You can't change $\mathcal{E}$ and r, of course.)

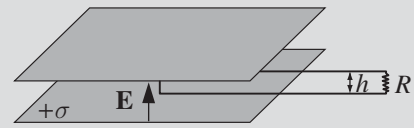


Fig. 7.9

Problem 7.6. A rectangular loop of wire is situated so that one end (height h) is between the plates of a parallel-plate capacitor (Fig. 7.9), oriented parallel to the field $\mathbf{E}$. The other end is way outside, where the field is essentially zero. What is the emf in this loop? If the total resistance is R, what current flows? Explain. [*Warning:* This is a trick question, so be careful; if you have invented a perpetual motion machine, there's probably something wrong with it.]

7.1.3 Motional Emf

In the last section, I listed several possible sources of electromotive force, batteries being the most familiar. But I did not mention the commonest one

[4] *Real* batteries have a certain **internal resistance**, r, and the potential difference between their terminals is $\mathcal{E} - Ir$, when a current I is flowing. For an illuminating discussion of how batteries work, see D. Roberts, *Am. J. Phys.* **51**, 829 (1983).

[5] Current in an electric circuit is somewhat analogous to the flow of water in a closed system of pipes, with gravity playing the role of the electrostatic field, and a pump (lifting the water up *against* gravity) in the role of the battery. In this story, *height* is analogous to voltage.

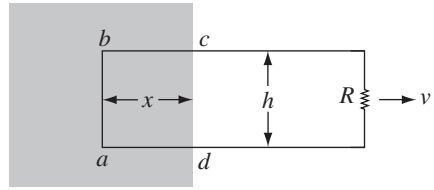


Fig. 7.10

of all: the **generator**. Generators exploit **motional emfs**, which arise when you *move a wire through a magnetic field*. Figure 7.10 suggests a primitive model for a generator. In the shaded region there is a uniform magnetic field **B**, pointing into the page, and the resistor R represents whatever it is (maybe a light bulb or a toaster) we're trying to drive current through. If the entire loop is pulled to the right with speed v, the mobile charges in segment ab experience a magnetic force whose vertical component qvB drives current around the loop, in the clockwise direction. The emf is

$$\mathcal{E} = \oint \mathbf{f}_{\text{mag}} \cdot d\mathbf{l} = vBh, \tag{7.11}$$

where h is the width of the loop. (The horizontal segments bc and ad contribute nothing, since the force there is perpendicular to the wire.)

Notice that the integral you perform to calculate $\mathcal{E}$ (Eq. 7.9 or 7.11) is carried out at *one instant of time* – take a "snapshot" of the loop, if you like, and work from that. Thus $d\mathbf{l}$, for the segment ab in Fig. 7.10, points straight up, even though the loop is moving to the right. You can't quarrel with this – it's simply the way emf is *defined* – but it is important to be clear about it.

In particular, although the magnetic force is responsible for establishing the emf, it is *not* doing any work – magnetic forces *never* do work. Who, then, *is* supplying the energy that heats the resistor? *Answer:* The person who's pulling on the loop, of course. But this is subtle, so listen closely. With the current flowing, the free charges in segment ab have a vertical velocity (call it **u**) in addition to the horizontal velocity **v** they inherit from the motion of the loop. Accordingly, the magnetic force has a component quB to the left. To counteract this, the person pulling on the wire must exert a force per unit charge

$$f_{\text{pull}} = uB$$

to the *right* (Fig. 7.11). This force is transmitted to the charge by the structure of the wire.

Meanwhile, the particle is actually *moving* in the direction of the resultant velocity **w**, and the distance it goes is $(h/\cos\theta)$. The work done per unit charge (*by* the person pulling on the loop, *on* a charge as it ascends the "vertical" segment) is therefore

$$\int \mathbf{f}_{\text{pull}} \cdot d\mathbf{l} = (uB)\left(\frac{h}{\cos\theta}\right)\sin\theta = vBh = \mathcal{E}$$

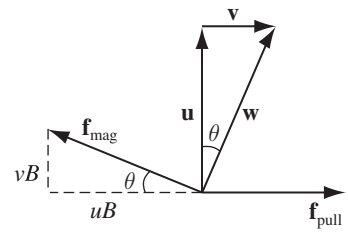
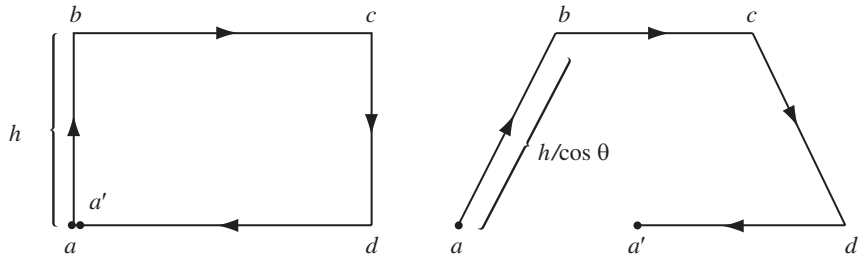


Fig. 7.11



(a) Integration path for computing $\mathcal{E}$ (follow the wire at one instant of time).

(b) Integration path for calculating work done (follow the charge around the loop).

Fig. 7.12

(sin θ coming from the dot product). As it turns out, then, the *work done per unit charge is exactly equal to the emf*, though the integrals are taken along entirely different paths (Fig. 7.12), and completely different forces are involved. To calculate the emf, you integrate around the loop at *one instant*, but to calculate the work done you follow a charge in its journey around the loop; $\mathbf{f}_{\text{pull}}$ contributes nothing to the emf, because it is perpendicular to the wire, whereas $\mathbf{f}_{\text{mag}}$ contributes nothing to work because it is perpendicular to the net motion of the charge.[6]

There is a particularly nice way of expressing the emf generated in a moving loop. Let Φ be the flux of $\mathbf{B}$ through the loop:

$$\Phi \equiv \int \mathbf{B} \cdot d\mathbf{a}. \tag{7.12}$$

For the rectangular loop in Fig. 7.10,

$$\Phi = Bhx.$$

As the loop moves, the flux decreases:

$$\frac{d\Phi}{dt} = Bh\frac{dx}{dt} = -Bhv.$$

(The minus sign accounts for the fact that dx/dt is negative.) But this is precisely the emf (Eq. 7.11); the emf generated in the loop is minus the rate of change of flux through the loop:

[6] For further discussion, see E. P. Mosca, *Am. J. Phys.* **42**, 295 (1974).

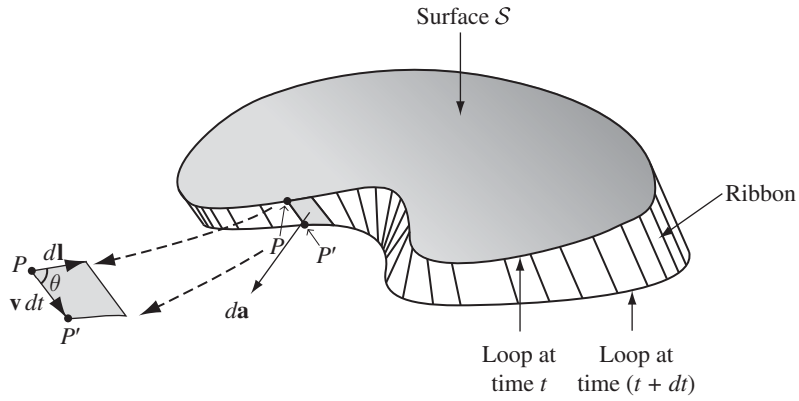


Fig. 7.13

$$\boxed{\mathcal{E} = -\frac{d\Phi}{dt}.} \tag{7.13}$$

This is the **flux rule** for motional emf.

Apart from its delightful simplicity, the flux rule has the virtue of applying to *non*rectangular loops moving in *arbitrary* directions through *non*uniform magnetic fields; in fact, the loop need not even maintain a fixed shape.

Proof. Figure 7.13 shows a loop of wire at time t, and also a short time dt later. Suppose we compute the flux at time t, using surface $\mathcal{S}$, and the flux at time $t + dt$, using the surface consisting of $\mathcal{S}$ plus the "ribbon" that connects the new position of the loop to the old. The *change* in flux, then, is

$$d\Phi = \Phi(t + dt) - \Phi(t) = \Phi_{\text{ribbon}} = \int_{\text{ribbon}} \mathbf{B} \cdot d\mathbf{a}.$$

Focus your attention on point P: in time dt, it moves to P'. Let $\mathbf{v}$ be the velocity of the *wire*, and $\mathbf{u}$ the velocity of a charge *down* the wire; $\mathbf{w} = \mathbf{v} + \mathbf{u}$ is the resultant velocity of a charge at P. The infinitesimal element of area on the ribbon can be written as

$$d\mathbf{a} = (\mathbf{v} \times d\mathbf{l})\, dt$$

(see inset in Fig. 7.13). Therefore

$$\frac{d\Phi}{dt} = \oint \mathbf{B} \cdot (\mathbf{v} \times d\mathbf{l}).$$

Since $\mathbf{w} = (\mathbf{v} + \mathbf{u})$ and $\mathbf{u}$ is parallel to $d\mathbf{l}$, we can just as well write this as

$$\frac{d\Phi}{dt} = \oint \mathbf{B} \cdot (\mathbf{w} \times d\mathbf{l}).$$

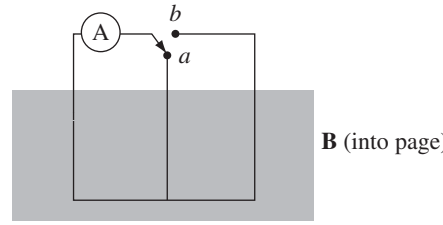


Fig. 7.14

Now, the scalar triple product can be rewritten:

$$\mathbf{B} \cdot (\mathbf{w} \times d\mathbf{l}) = -(\mathbf{w} \times \mathbf{B}) \cdot d\mathbf{l},$$

so

$$\frac{d\Phi}{dt} = -\oint (\mathbf{w} \times \mathbf{B}) \cdot d\mathbf{l}.$$

But $(\mathbf{w} \times \mathbf{B})$ is the magnetic force per unit charge, $\mathbf{f}_{\text{mag}}$, so

$$\frac{d\Phi}{dt} = -\oint \mathbf{f}_{\text{mag}} \cdot d\mathbf{l},$$

and the integral of $\mathbf{f}_{\text{mag}}$ is the emf:

$$\mathcal{E} = -\frac{d\Phi}{dt}. \qquad \square$$

There is a sign ambiguity in the definition of emf (Eq. 7.9): Which *way* around the loop are you supposed to integrate? There is a compensating ambiguity in the definition of *flux* (Eq. 7.12): Which is the positive direction for $d\mathbf{a}$? In applying the flux rule, sign consistency is governed (as always) by your right hand: if your fingers define the positive direction around the loop, then your thumb indicates the direction of $d\mathbf{a}$. Should the emf come out negative, it means the current will flow in the negative direction around the circuit.

The flux rule is a nifty shortcut for calculating motional emfs. It does not contain any new physics – just the Lorentz force law. But it can lead to error or ambiguity if you're not careful. The flux rule assumes you have a single wire loop – it can move, rotate, stretch, or distort (continuously), but beware of switches, sliding contacts, or extended conductors allowing a variety of current paths. A standard "flux rule paradox" involves the circuit in Fig. 7.14. When the switch is thrown (from a to b) the flux through the circuit doubles, but there's no motional emf (no conductor moving through a magnetic field), and the ammeter (A) records no current.

Example 7.4. A metal disk of radius a rotates with angular velocity ω about a vertical axis, through a uniform field $\mathbf{B}$, pointing up. A circuit is made by connecting one end of a resistor to the axle and the other end to a sliding contact, which touches the outer edge of the disk (Fig. 7.15). Find the current in the resistor.

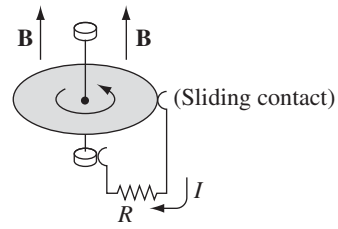


Fig. 7.15

Solution. The speed of a point on the disk at a distance s from the axis is $v = \omega s$, so the force per unit charge is $\mathbf{f}_{\text{mag}} = \mathbf{v} \times \mathbf{B} = \omega s B \hat{\mathbf{s}}$. The emf is therefore

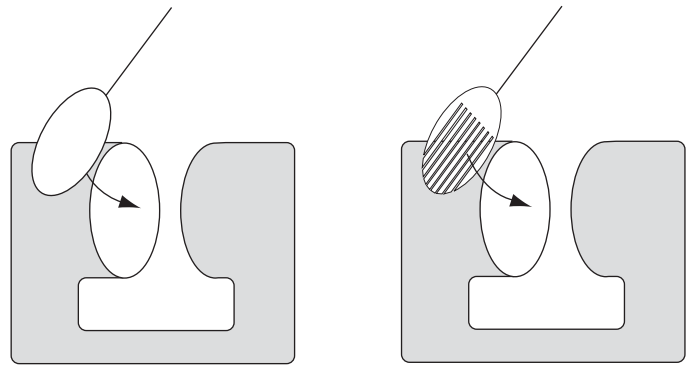
$$\mathcal{E} = \int_0^a f_{\text{mag}}\, ds = \omega B \int_0^a s\, ds = \frac{\omega B a^2}{2},$$

and the current is

$$I = \frac{\mathcal{E}}{R} = \frac{\omega B a^2}{2R}.$$

Example 7.4 (the **Faraday disk**, or **Faraday dynamo**) involves a motional emf that you can't calculate (at least, not directly) from the flux rule. The flux rule assumes the current flows along a well-defined path, whereas in this example the current spreads out over the whole disk. It's not even clear what the "flux through the circuit" would *mean* in this context.

Even more tricky is the case of **eddy currents**. Take a chunk of aluminum (say), and shake it around in a nonuniform magnetic field. Currents will be generated in the material, and you will feel a kind of "viscous drag" – as though you were pulling the block through molasses (this is the force I called $\mathbf{f}_{\text{pull}}$ in the discussion of motional emf). Eddy currents are notoriously difficult to calculate,[7] but easy and dramatic to demonstrate. You may have witnessed the classic experiment in which an aluminum disk mounted as a



(a) (b)

Fig. 7.16

[7] See, for example, W. M. Saslow, *Am. J. Phys.* **60**, 693 (1992).

pendulum on a horizontal axis swings down and passes between the poles of a magnet (Fig. 7.16a). When it enters the field region it suddenly slows way down. To confirm that eddy currents are responsible, one repeats the demonstration using a disk that has many slots cut in it, to prevent the flow of large-scale currents (Fig. 7.16b). This time the disk swings freely, unimpeded by the field.

Example 7.5.[8] There are precious few eddy current problems that can be treated analytically. Here's a rare example, solved by Hertz (at the age of 23) in 1880. Imagine a uniform, electrically neutral, spherical shell, of inner radius R_1, outer radius R_2, and conductivity σ, centered at the origin and spinning with constant angular velocity $\boldsymbol{\omega}$, in the presence of a uniform magnetic field $\mathbf{B}$.[9] Find the eddy currents in the shell, in the steady state (fields, charges, currents all constant in time), if (a) $\boldsymbol{\omega}$ and $\mathbf{B}$ are in the *same* direction and (b) they are perpendicular.

Solution. In electrostatics the charge density in a conductor is zero (any excess charge goes to the surface), and this is still true for steady currents in conductors at rest, as we found in Eq. 7.5. But for a conductor *moving* through a magnetic field we cannot ignore the velocity term in Eq. 7.2. Indeed, taking the divergence, and invoking Gauss's law and the continuity equation,

$$\boldsymbol{\nabla}\cdot\mathbf{J} = \sigma[\boldsymbol{\nabla}\cdot\mathbf{E} + \boldsymbol{\nabla}\cdot(\mathbf{v}\times\mathbf{B})] = \sigma\Big[\frac{1}{\epsilon_0}\rho + \boldsymbol{\nabla}\cdot(\mathbf{v}\times\mathbf{B})\Big] = -\frac{\partial\rho}{\partial t}.$$

In the steady state ρ is constant, and we are left with

$$\rho = -\epsilon_0\boldsymbol{\nabla}\cdot(\mathbf{v}\times\mathbf{B}). \tag{7.14}$$

(a) We might as well choose axes such that $\mathbf{B} = B\hat{\mathbf{z}}$ and $\boldsymbol{\omega} = \omega\hat{\mathbf{z}}$. For any point in a rigid rotating object, $\mathbf{v} = \boldsymbol{\omega}\times\mathbf{r}$, so, using the BAC–CAB rule,

$$\mathbf{v}\times\mathbf{B} = (\boldsymbol{\omega}\times\mathbf{r})\times\mathbf{B} = \mathbf{r}(\mathbf{B}\cdot\boldsymbol{\omega}) - \boldsymbol{\omega}(\mathbf{B}\cdot\mathbf{r}) = \omega B(\mathbf{r} - z\hat{\mathbf{z}}) = \omega B(x\hat{\mathbf{x}} + y\hat{\mathbf{y}}), \tag{7.15}$$

and therefore

$$\rho = -2\epsilon_0\omega B. \tag{7.16}$$

(There will be compensating charge at the two surfaces, of course.) The electric potential satisfies Poisson's equation

$$\nabla^2 V = -\frac{1}{\epsilon_0}\rho = 2\omega B \quad (R_1 < r < R_2) \tag{7.17}$$

[8] This example can be skipped without loss of continuity. For details and practical applications see M. A. Nurge et al., *Am. J. Phys.* **85**, 596 (2017).

[9] Assume that any magnetic fields produced by the eddy currents themselves are negligible – or that they are included in $\mathbf{B}$.

and the radial current vanishes at the boundaries (otherwise charge would be leaking off into space):

$$J_r = \sigma[\mathbf{E} + \mathbf{v} \times \mathbf{B}] \cdot \hat{\mathbf{r}} = 0 \quad (\text{at } r = R_1 \text{ and } R_2). \tag{7.18}$$

Thus

$$E_r = -\omega B(x\hat{\mathbf{x}} + y\hat{\mathbf{y}}) \cdot \hat{\mathbf{r}} = -\frac{\omega B}{r}(x^2 + y^2) = -\omega B r \sin^2\theta \tag{7.19}$$

at each surface:

$$\frac{\partial V}{\partial r}\Big|_{R_1} = \omega B R_1 \sin^2\theta, \quad \frac{\partial V}{\partial r}\Big|_{R_2} = \omega B R_2 \sin^2\theta. \tag{7.20}$$

Having specified the charge density (Eq. 7.16) and the normal derivative of V on the boundaries (Eq. 7.20), the field in the shell is uniquely determined (Prob. 3.5). In fact, it's easy to guess the potential:

$$V(r, \theta) = \frac{\omega B}{2} r^2 \sin^2\theta + K = \frac{\omega B}{2}(x^2 + y^2) + K \tag{7.21}$$

(for some constant K), and the field is

$$\mathbf{E} = -\boldsymbol{\nabla} V = -\omega B(x\hat{\mathbf{x}} + y\hat{\mathbf{y}}) = -\mathbf{v} \times \mathbf{B} \tag{7.22}$$

(Eq. 7.15). As it turns out, then, the electric force exactly cancels the magnetic force; there is *no* eddy current in this configuration,[10]

$$\mathbf{J} = \mathbf{0}.$$

(b) This time we'll take $\boldsymbol{\omega} = \omega\hat{\mathbf{x}}$ and $\mathbf{B} = B\hat{\mathbf{y}}$. Then

$$\mathbf{v} \times \mathbf{B} = \mathbf{r}(\mathbf{B} \cdot \boldsymbol{\omega}) - \boldsymbol{\omega}(\mathbf{B} \cdot \mathbf{r}) = -\omega B y \hat{\mathbf{x}},$$

which is divergenceless, so the charge density vanishes (Eq. 7.14):

$$\rho = 0$$

(there may still be induced surface charges). Laplace's equation holds within the shell,

$$\nabla^2 V = 0,$$

and Eq. 7.19 becomes

$$E_r = -(\mathbf{v} \times \mathbf{B}) \cdot \hat{\mathbf{r}} = \omega B y(\hat{\mathbf{x}} \cdot \hat{\mathbf{r}}) = \omega B r \sin^2\theta \sin\phi \cos\phi$$

at each boundary:

$$\frac{\partial V}{\partial r}\Big|_{R_1} = -\omega B R_1 \sin^2\theta \sin\phi \cos\phi, \frac{\partial V}{\partial r}\Big|_{R_2} = -\omega B R_2 \sin^2\theta \sin\phi \cos\phi.$$

The solution is

$$V(r, \theta, \phi) = -\frac{\omega B}{2} r^2 \sin^2\theta \sin\phi \cos\phi + K = -\frac{\omega B}{2} xy + K (R_1 < r < R_2),$$

[10] There is a **convection current**, since ρ (and the surface charges) are dragged around by the spinning of the shell, but (as you can check for yourself) the ratio $J_{\text{conv}}/J_{\text{cond}}$ is typically of the order of $\epsilon_0\omega/\sigma$, which is utterly negligible for (say) aluminum at 100 rpm.

and the electric field is

$$\mathbf{E} = -\boldsymbol{\nabla} V = \frac{\omega B}{2}(y\hat{\mathbf{x}} + x\hat{\mathbf{y}}).$$

The eddy current is

$$\mathbf{J} = \sigma[\mathbf{E} + (\mathbf{v} \times \mathbf{B})] = \frac{\omega B\sigma}{2}(-y\hat{\mathbf{x}} + x\hat{\mathbf{y}}) = \frac{\omega B\sigma}{2} r \sin\theta\,\hat{\boldsymbol{\phi}}.$$

Remarkably, this is just what you would get if the shell had been a *non*conductor carrying a uniform charge density $\rho = B\sigma/2$ and spinning with angular velocity ω about the z-axis. That's *right*: the z-axis! Incidentally, for $\boldsymbol{\omega}$ and $\mathbf{B}$ in arbitrary orientations the formula is

$$\mathbf{J} = -\frac{\sigma}{2}[\mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{B})].$$

Problem 7.7. A metal bar of mass m slides frictionlessly on two parallel conducting rails a distance l apart (Fig. 7.17). A resistor R is connected across the rails, and a uniform magnetic field $\mathbf{B}$, pointing into the page, fills the entire region.

(a) If the bar moves to the right at speed v, what is the current in the resistor? In what direction does it flow?
(b) What is the magnetic force on the bar? In what direction?
(c) If the bar starts out with speed v_0 at time $t = 0$, and is left to slide, what is its speed at a later time t?
(d) The initial kinetic energy of the bar was, of course, $\frac{1}{2}mv_0{}^2$. Check that the energy delivered to the resistor is exactly $\frac{1}{2}mv_0{}^2$.

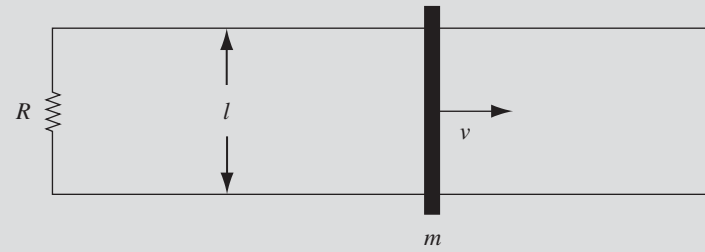


Fig. 7.17

Problem 7.8. A square loop of wire (side length a) lies on a table, a distance s from a very long straight wire, which carries a current I, as shown in Fig. 7.18.

(a) Find the flux of $\mathbf{B}$ through the loop.
(b) If someone now pulls the loop directly away from the wire, at speed v, what emf is generated? In what direction (clockwise or counterclockwise) does the current flow?
(c) What if the loop is pulled to the *right* at speed v?

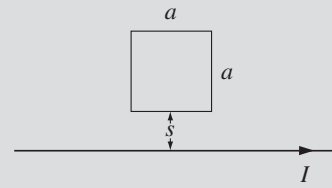


Fig. 7.18

Problem 7.9. An infinite number of different surfaces can be fit to a given boundary line, and yet, in defining the magnetic flux through a loop, $\Phi = \int \mathbf{B} \cdot d\mathbf{a}$, I never specified the particular surface to be used. Justify this apparent oversight.

Problem 7.10. A square loop (side length a) is mounted on a vertical shaft and rotated at angular velocity ω (Fig. 7.19). A uniform magnetic field $\mathbf{B}$ points to the right. Find $\mathcal{E}(t)$ for this **alternating current** generator.

Problem 7.11. A square loop is cut out of a thick sheet of aluminum. It is then placed so that the top portion is in a uniform magnetic field $\mathbf{B}$, and is allowed to fall under gravity (Fig. 7.20). (In the diagram, shading indicates the field region; $\mathbf{B}$ points into the page.) If the magnetic field is 1 T (a pretty standard laboratory field), find the terminal velocity of the loop (in m/s). Find the velocity of the loop as a function of time. How long does it take (in seconds) to reach, say, 90 percent of the terminal velocity? What would happen if you cut a tiny slit in the ring, breaking the circuit? [*Note:* The dimensions of the loop cancel out; determine the actual *numbers*, in the units indicated.]

Problem 7.12. Calculate the net magnetic torque on the eddy current in Ex. 7.5b. This torque will act as a brake on the rotation. What is the time constant for the (exponential) damping? Assume that the angular velocity lost in a given rotation is a small fraction of the total. Find the actual number, for aluminum (mass density 2.7×10^3 kg/m^3) in the Earth's magnetic field.

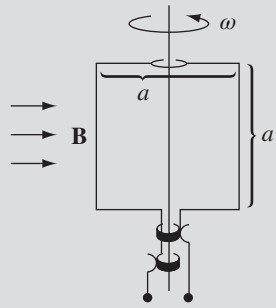


Fig. 7.19

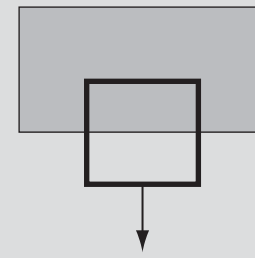


Fig. 7.20